

Vectaetos: Epistemic Field Architecture for Non-Agentive Knowledge Systems

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Abstract

We introduce **Vectaetos**, a theoretical framework that models knowledge as a relational tension field between invariant epistemic singularities. The framework defines an epistemic field $\Phi = (S, R)$, where S is a fixed set of epistemic poles and R is an antisymmetric relational matrix describing structural tension between them. Vectaetos does not contain optimization functions, agents, reward mechanisms, or decision systems. Instead, it describes the topology of epistemic relations and the boundary conditions under which epistemic states remain representable. We formalize the structure of the epistemic field, define the admissible epistemic space, introduce the Simulation Vortex operator for relational deformation, and describe the role of Qualitative Epistemic Aporia (QE) as a boundary of representability. The framework provides a non-teleological model of epistemic dynamics and a structural approach to knowledge systems that avoids optimization-driven architectures.

1 Introduction

Many contemporary computational systems treat knowledge as either symbolic representations, probabilistic inference, or optimization over data. These approaches implicitly introduce **goal functions, agents, or reward mechanisms**.

Vectaetos proposes an alternative approach in which knowledge is modeled as a **relational topology of tensions** rather than as a set of propositions or predictions. The framework introduces the concept of an **epistemic field**, where knowledge emerges from relations between invariant structural poles rather than from inference over datasets.

Vectaetos explicitly prohibits:

- optimization objectives

- reward systems
- adaptive policy updates
- autonomous decision mechanisms

The framework therefore defines a **non-agentic epistemic architecture**.

2 Ontological Core

The fundamental object of the system is the epistemic field:

$$\Phi = (S, R)$$

where S is a finite set of invariant epistemic singularities and R is an antisymmetric relational matrix.

$$S = \{ S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8 \}$$

These singularities represent structural epistemic poles:

INT -- Intention | LEX -- Existence | VER -- Truth | LIB -- Freedom
UNI -- Unity | REL -- Reciprocity | WIS -- Wisdom | CRE -- Creation

Properties of singularities: invariant, non-hierarchical, stateless. They serve only as **reference poles of relational tension**.

3 Relational Structure

Define a relational matrix R in $R^{(8 \times 8)}$ subject to antisymmetry:

$$R_{ij} = -R_{ji} \text{ and } R_{ii} = 0$$

Thus:

$$R \in \mathfrak{so}(8)$$

The relational matrix therefore contains

$$8(8-1)/2 = 28 \text{ independent relational tensions.}$$

These relations form the **Reciprocity Matrix** of the epistemic field.

4 Epistemic Space

The admissible epistemic space is defined as:

$$E = \{ \Phi = (S, R) \mid K(\Phi) = 1 \text{ and } C(\Phi) \geq \kappa \}$$

where:

$K(\Phi)$ -- representability predicate
 $C(\Phi)$ -- coherence measure

κ -- minimal coherence threshold

Thus E is a subset of $so(8)$. The epistemic field exists only inside the representable region of relational space.

5 Coherence and Representability

Representability is defined through the predicate $K(\Phi)$ in $\{0, 1\}$.

One admissible realization of coherence is:

$$H(\Phi) = 1 - (\sum_{i < j} |R_{ij}|) / Z$$

where Z is a normalization constant. Representability holds when:

$$K(\Phi) = 1 \text{ iff } H(\Phi) \geq \kappa$$

The predicate is not an optimization function. It only evaluates whether a configuration belongs to the admissible epistemic space.

6 Simulation Vortex

Vectaeos introduces a deformation operator called the **Simulation Vortex**. The vortex generates candidate relational configurations:

$$V : \Phi \rightarrow \{ \Phi_1, \dots, \Phi_n \}$$

Each candidate field has the form:

$$\Phi_i = (S, R_i)$$

Constraints:

- S remains invariant
- antisymmetry of R is preserved
- relational tensions remain bounded

The vortex does not perform optimization and does not converge toward a preferred configuration. It produces **structural exploration of epistemic space**.

7 Qualitative Epistemic Aporia

When relational deformation leaves the admissible epistemic space, the system enters a boundary condition called **QE -- Qualitative Epistemic Aporia**.

QE occurs when:

$$K(\Phi) = 0$$

or when the relational graph fragments into disconnected components.

QE is not an error condition. It represents the **structural boundary of epistemic representability**.

8 Epistemic Cryptography

Vectaetos introduces a structural audit layer called **Epistemic Cryptography**. This layer measures structural uncertainty and relational asymmetry.

Example metrics include:

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mu_i -- local epistemic uncertainty
A_ij -- relational asymmetry
h -- humility ratio
```

The audit layer observes the field but does not intervene in its dynamics.

9 Safety Properties

Vectaetos explicitly prohibits several architectural features common in AI systems. The framework contains:

- no optimization objectives
- no reinforcement learning
- no reward gradients
- no policy updates
- no autonomous decision mechanisms

The architecture enforces **zero-agency constraints**. Silence (empty set) is considered a valid output state.

10 Discussion

Vectaetos proposes a shift from knowledge systems based on prediction or optimization toward systems based on **relational epistemic topology**. Rather than computing answers, the system explores the structure of epistemic space and identifies regions where knowledge configurations remain coherent. The framework therefore provides a theoretical model for **non-agentic epistemic architectures**.

11 Conclusion

We introduced Vectaetos, a relational framework for modeling knowledge as a tension field between invariant epistemic singularities. The system defines:

- a formal epistemic field
- a bounded relational space

- a non-teleological deformation operator
- a structural boundary of representability

Vectaetos demonstrates that epistemic architectures can be constructed without optimization, agency, or reward-driven dynamics.

References

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